rogii-wellbore-geology-prediction

21st Place Solution

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Team	goonew
Author	goonew
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I would like to thank the organisers for this valuable opportunity, as well as everyone who shared their knowledge on the discussion forum.

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1. Overview of the Solution

The most effective component of my solution was synthetic-data generation. I began with several findings that had already been established on the discussion forum early in the competition.

- I assumed that the labels were generated from the sparse dip annotations produced by StarSteer, Rogii's geosteering software. Hengck23 pointed this out in discussion 699853 ("annotation leak! that is how starsteer works"). In the same thread, Tom reported that `ANCC` (the formation boundary) is piecewise linear, with about 15 control points per well—roughly one every 323 rows. I independently confirmed that $S = Z + TVT$ (the elevation of the structural surface) can be reproduced by a piecewise-linear function with the same number of nodes, with a residual standard deviation of 0.88 ft.
- Hengck23 also noted early on that StarSteer's search procedure is described in patents as a beam/Viterbi search over dip ranges and large segments (discussion 699289, discussion 699883). Based on this, I assumed a greedy search in which the main objective is the segment-wise Pearson correlation between the horizontal-well GR and a synthetic log obtained by resampling the type-well log along a candidate dip. I added penalties for dip differences between adjacent segments and for deviations from the regional dip, and used this formulation as the specification for synthetic-data generation.
- In my own oracle measurements, a single dip—that is, a straight line—gave an RMSE of 7.93, while a two-segment model gave 4.97.

Based on these findings, I trained on synthetic data designed to imitate the ground-truth generator and decoded TVT using a structured decoder. Here, "decoder" does not mean a regression model that independently predicts one value per row. It refers to an inference procedure that selects the TVT sequence that best aligns the GR with the type-well log across the entire well, subject to the structural constraints that the dip remains constant within each segment and that changes are sparse.

Pipeline

I reproduced a public notebook, strengthened it, and then applied a sequence of decoder-based corrections on top of the resulting baseline.

CV was measured using well-grouped 5-fold cross-validation with two seeds (42 and 2026). I report four metrics, defined as follows.

- **Pooled RMSE**: the standard RMSE calculated over all evaluation rows pooled together.
- **Iso (iso155) RMSE**: pooled RMSE restricted to the 155 wells with the fewest neighbouring wells.
- **Median RMSE**: the median of the per-well RMSE values.
- **Crit5 RMSE**: pooled RMSE restricted to five wells—1b1eba53, 91b301ce, 91db7070, f2d4c8c9, and fb03ae90—which had the largest squared errors.

#	Stage	Main mechanism	CV pooled	CV iso	CV median	CV crit 5	public	private
0	Reproducing a public notebook	GR tracking with a particle filter	10.988	12.609	5.779	39.340	8.988	10.763
1	Strengthening the baseline	Combined neighbour-well transfer, piecewise-linear replay decoder, and particle filter	6.511	7.171	3.815	25.907	6.723	7.506
2	Candidate scorer trained on synthetic data	Candidate-selection CNN trained only on synthetic wells that imitate the generator	6.331	6.979	3.497	26.204	6.436	7.049
3	Cost volume + forward–backward	Learning likelihoods on a row × offset lattice and decoding with the posterior mean	5.999	6.878	3.210	26.043	6.397	6.815
4	Diversifying the reference	A reference-free absolute decoder, plus the well's own known interval as a second reference log	5.189	6.025	2.818	23.775	5.649	6.557
5	Recombination layer (final submission)	Recombining the contribution of each stage, conditioned on uncertainty	5.028	5.809	2.799	22.161	5.587	6.504

Adoption Criteria

In BirdCLEF 2026, the gap between the public and private leaderboards happened to be small, which helped me win a gold medal. In this competition, I again tried to keep that gap as small as possible. Rather than deciding whether to adopt an idea based only on pooled CV RMSE, I also examined the median per-well RMSE and the fraction of wells that improved, so that I could verify that an improvement benefited a broad set of wells. I rejected any idea that worsened the median, even when it improved pooled CV. In other words, I rejected improvements that substantially fixed a small number of difficult wells at the cost of slightly worsening performance on the majority of wells.

In fact, a model with a better pooled CV score of 4.80 achieved only 6.57 on the private leaderboard. Had I trusted pooled CV alone, I would not have won a medal. That said, the final public–private gap was still about 1 ft, so the outcome may simply have involved some luck.

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3. Stage 1: Strengthening the Baseline

The strongest public notebook used a particle filter whose state consisted of $S = TVT + Z$ and a velocity term. It tracked the horizontal well by using agreement with the type-well GR profile as the likelihood, and achieved a public leaderboard score of roughly 9.15. By that point, several standard patterns had also been established in public notebooks and forum discussions: predicting residuals, NCC matching against the type-well GR, and Ridge/NNLS stacking of multiple methods. I reproduced these ideas and then extended them as follows.

- **Switching to residual regression**: instead of predicting the raw TVT, which is around 11,000 ft, predict the drift from the end of the known interval, which is roughly ±16 ft. This idea came from public notebooks.
- **Pairwise transfer from neighbouring wells**: the use of neighbouring wells as spatial context had already been proposed on the forum, for example through the spatial-neighbourhood consensus described in

discussion 699289. My change was in how that information was transferred. After removing a constant offset, the shape of S agrees across neighbouring wells to within a MAD of 3–7 ft. Rather than fitting all wells as a single spatial field, I aligned each neighbouring well directly to the target well's known interval, removed the offset, and only then transferred its information. This reduced the standalone CV from 21.7 to 12.2. The neighbour bank was constructed in a leave-one-well-out manner.

- **Piecewise-linear decoder**: I implemented a decoder that re-ran the search process I assumed had generated the labels. It explicitly enforced the constraints "constant dip within a segment" and "sparse switches", and searched for the piecewise-linear path that maximised segment-wise GR agreement. Instead of selecting the single highest-posterior path, I used the posterior mean from the forward–backward algorithm—the standard exact inference procedure for hidden Markov models, which computes the marginal posterior at each row by dynamic programming. This reduced the heavy tail of the between-well error distribution and improved RMSE by 0.37. The reference curve had to preserve the dip structure faithfully rather than merely achieve a low RMSE; an over-smoothed reference flattened the dip prior and made performance worse.

Combining these components gave CV 6.51, public 6.723, and private 7.506.

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4. Stage 2: A Candidate Scorer Trained on Synthetic Data

As described above, I assumed a generative process for the ground truth and estimated its parameters directly from the training data. The dip increments followed a first-order autoregressive process ($\phi = 0.636$ per 100 ft), with Brownian noise, a correlation length of 300–500 ft, and no periodicity. The trajectory steering and GR residual were also first-order autoregressive, while the misalignment between the type well and horizontal well followed an Ornstein–Uhlenbeck process.

I calibrated the difficulty of the synthetic wells against the real data using two statistics. The first was the number of local minima in the matching-cost curve. When the candidate TVT offset is scanned over a range of several tens of feet, the mismatch between the horizontal-well GR and the type-well log forms a curve with many valleys. The number of valleys reflects the number of plausible solutions for that well, and therefore the degree of aliasing. The synthetic data produced 9.1 local minima per well, compared with 10.1 in the real data.

The second statistic was the proportion of cases in which the minimum of that curve coincided with the ground truth. This measures how much of the problem can be solved through naive matching alone. The values were 0.855 for synthetic data and 0.863 for real data.

Using this process, I could generate a virtually unlimited number of synthetic wells with ground-truth labels.

What I Trained the Model to Do

I trained the model to score candidates rather than perform regression. Around the base prediction, I generated many candidate paths parameterised by dip a and translation b. A 1-D CNN operating on seven channels—including normalised GR, the type-well log resampled along the candidate dip, a mask, trajectory dip, and local misfit—assigned a score to each candidate. The final prediction was the softmax posterior mean.

Maintaining a resolution of 1 ft was essential. Decimation or smoothing destroyed the GR correlation structure, whose characteristic length is about 3 ft, and the usable signal disappeared.

I avoided regression because the error distribution was bimodal: a prediction was typically either correct or displaced by one bundle. Souldrive explained the origin of this bimodality in discussion 711878: because of the cyclic bedding in the Eagle Ford, the GR can match equally well at two horizons separated by approximately 15 ft. A regression model inevitably hedges between the two modes and loses information. A candidate-scoring model, by contrast, can in principle select one of the modes.

Scaling the model, ensembling several models, and applying the entire procedure twice reduced CV from 6.51 to 6.30. This layer also transferred directly to the leaderboard, improving the private score from 7.506 to 7.049, and became the foundation of every subsequent submission.

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5. Stage 3: Cost-Volume Likelihoods and Forward–Backward

The Stage 2 scorer used a single softmax over the entire well. In this stage, I replaced that with row-level decoding at 1 ft resolution.

I used a structure similar to the cost volume employed in stereo matching, the classical problem of estimating depth from a pair of images. In stereo matching, one constructs an array containing the matching cost for every pixel and every disparity candidate, then extracts the most consistent depth surface. Here, I constructed a lattice over each row and each offset δ from the reference. An 8-channel 2-D CNN emitted a log-likelihood at every lattice point. Training used the same synthetic-data factory as Stage 2, together with injected calibration errors.

The values on this lattice correspond to the emissions in a hidden Markov model: if the true position at row i were offset by δ, how plausible would the observed GR be? For the transition probabilities, I directly used the measured dip-increment process—an OU process with a heavy tail—and decoded the posterior mean using exact forward–backward inference.

Applied naively, this approach performed worse on real data. The reason was clear: forward–backward effectively compounds roughly 600 correlated emissions along a path, so even mild overconfidence is amplified. Stage 2 used only a single softmax per well and therefore did not suffer from the same effect. I introduced two countermeasures.

- **Emission decimation and an entropy gate**: I decimated the emissions to one every 32 ft and smoothly reduced the contribution of high-entropy positions towards zero. This worked substantially better than applying a uniform temperature adjustment. The entropy was also well calibrated on real data: among the sharpest 20% of positions, 89% had errors within 1.4 ft, whereas the corresponding figures for the flattest 20% were 39% and 7.4 ft.
- **Fine-tuning on real data**: I took a network pre-trained exclusively on synthetic data and briefly fine-tuned it on the real data within each fold, using 2,000 steps and a low learning rate. Every well was therefore evaluated using a network trained on a fold that excluded that well. Because this provided dense supervision over 470,000 points per fold, it was highly effective. Even under a pessimistic protocol in which the folds were redefined by region rather than by well, the discrepancy remained only +0.02, supporting the conclusion that the improvement came from genuine signal rather than neighbour leakage.

The benefit of larger-scale pre-training—12k → 120k → 240k steps—carried directly through to post-fine-tuning performance and remained consistent until the end. This stage alone reduced CV from 6.30 to 5.60, the largest improvement produced by any single intervention.

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6. Stage 4: Diversifying the Reference

Up to this point, every decoder had operated as a $\pm\delta$ correction around an existing prediction, and most of the evidence came from matching GR against the type-well log. In this stage, I added two qualitatively different information sources.

The first was a reference-free absolute decoder. I built a Transformer that output, for each row, a probability distribution over the depth bins of the type-well log. Its output format was similar to that of a Pointer Network: rather than regressing a value, the model pointed to a position in the reference sequence. I supplied the `(GR, TVT)` pairs from the known interval as context tokens, and cross-attention placed each row of the evaluation zone directly onto the type well. This model was also pre-trained on synthetic data and fine-tuned on real data.

Its standalone RMSE of 14.18 was far worse than that of the full chain at the time, which was around 5.6. However, its errors were sufficiently independent that entropy-based per-well gating made it beneficial when merged into the ensemble.

The second source was the target well's own known interval, used as an additional reference log. I had explored the direction of using other wells as references extensively, but had not previously used the target well's own known interval as a reference curve. The TVT range of the known interval covered a median of 68% of the evaluation rows.

Reference	Correlation with the observed GR at the true position (median)
Type well	0.665
The well's own known interval	0.666
Blend of the two	0.725

On its own, the known interval performed about as well as the type well. Blending the two produced a substantial gain, indicating that they contained partially independent information. Furthermore, the correlation remained nearly constant with lateral distance from the heel—0.56–0.59 at both 0 ft and 4,500 ft. This suggests that the reference captures a stratigraphic signature that persists laterally, rather than merely local facies noise.

Rather than collapsing this information into a scalar feature, I injected it directly as an emission in the decoder. The state was defined as an offset of ± 8 ft from the current prediction. The emissions were multi-scale normalised cross-correlations over 500 ft and 800 ft windows, the transition was a Gaussian random walk, and only the shape component of the posterior mean was used in the final read-out.

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7. Stage 5: The Recombination Layer (Final Submission)

The final improvement from 5.19 to 5.028 came not from adding a new source of information, but from reusing information that had already been computed.

I decomposed the full chain into 18 "stage increments", each representing the correction added by one stage to the output of the preceding stage. After verifying that their sum reconstructed the submitted prediction exactly, with a maximum deviation of 1.8×10^{-12} ft, I multiplied each increment by a coefficient c. The squared error for well w can then be written as

$$ \mathrm{SSE}_w(c) = ee_w - 2\, c \cdot h_w + c^{\top} G_w c $$

where ee_w is the error obtained when all coefficients are zero, h_w is the inner product between each increment and the residual, and G_w is the Gram matrix of the increments. Once these three quantities had been computed for each well, the CV score of any coefficient vector could be evaluated almost instantly, allowing hundreds of configurations to be searched in a few seconds.

I then conditioned the level and shape components of each stage on uncertainty estimates produced by the chain itself, including the spread of the forward–backward posterior and the degree of disagreement within the ensemble. A control experiment clarified the role of this layer.

Design	pooled Δ	per-well median Δ	fraction of wells improved
One scalar per stage (unconditional)	−0.033	−0.002	0.505
Conditioned on uncertainty	−0.065	−0.023	0.567
Same degrees of freedom, covariates randomised	−0.024	+0.013	0.453

Even with randomised covariates, roughly one-third of the pooled gain was recovered. The improvement in the median, however, disappeared. In other words, conditioning did not merely increase the aggregate gain; it broadened the improvement across wells.

The final submission, was the first checkpoint from this layer. With the recombination layer itself refitted within each fold, it achieved a CV score of 5.028087, a median per-well ΔRMSE of −0.0245, and an improved-well fraction of 0.571. I continued developing this layer afterwards and reduced CV to 4.80, but the leaderboard score moved in the opposite direction.

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9. What Did Not Work for Me

Expanding the family of candidate paths did not help. Extending the candidate set all the way to the continuous least-squares solution reduced the oracle RMSE—assuming one could select the candidate closest to the ground truth—to 2.63, yet the actual scorer remained at 6.30. The entire gap came from candidate-selection quality, and the necessary information was not present in the observable signal because of the ± 1-bundle aliasing.

When the wells were divided into thirds according to the scorer's posterior variance, the most confident third had a level error of 2.51, essentially matching the oracle. For the flattest third, however, selecting the correct mode was effectively a coin flip, and the level error was 6.17. In that regime, the midpoint prediction is Bayes-optimal and cannot be improved further. This is the same limitation described by souldrive in discussion 711878.

10. Coding Agents

All ideas that materially improved CV were developed by me. In this project, LLMs did not generate useful modelling ideas or produce any major CV improvements. However, the coding agents consistently delivered fast implementations and detailed experiment reports, including HTML files with polished visualisations. This allowed me to spend more of my time analysing the problem and developing new ideas for improvement.